GGG-W-642B
August 8, 1968
SUPERSEDING
Fed. Spec. GGG-W-642A
April 9, 1963
(see 6.5)

FEDERAL SPECIFICATION

WRENCH, BUNG, (NONSPARKING AND NONMAGNETIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of bung wrench made of a nonferrous alloy having a low sparking hazard and a low magnetic permeability.

1.1.1 Federal specification coverage. Federal specifications do not include all types of the commodity indicated by the titles of the specifications, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

QQ-T-580 - Tool Steel, Carbon and Carbon-Vanadium.

Federal Standard:

Fed. Std. NO. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under general information in the index of Federal Specifications and Standards and at the prices indicated in the index. The index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-G-5572 - Gasoline, Aviation; Grades 80/87, 91/96, 100/130, 115/145.
MIL-H-15424 - Hand Tools; Packaging of.
MIL-I-17214 - Indicator, Permeability; Low-Mu (Go-No Go).

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification. Nonsparking, nonmagnetic bung wrenches furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Illustrations. The illustration herein is descriptive and not restrictive and is not intended to preclude the purchase of bung wrenches otherwise conforming to this specification.

3.3 Material. Nonsparking, nonmagnetic bung wrenches shall be made of a nonferrous-metal alloy or alloys, which will not cause an explosion when subjected to the sparking test specified in 4.6.3 and which shall have passed the magnetic permeability test specified in 4.6.4, and shall conform to the physical requirements hereinafter specified. In no case shall the iron content of the nonferrous alloy exceed 5 percent, and any iron content of the nonferrous alloy exceed 5 percent, and any iron content shall be well diffused in the alloy.

3.4 Marking. Each bung wrench shall be marked on the handle in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. Bung wrenches shall also be marked with the name, abbreviation, or chemical symbol of the alloy used in the fabrication of the wrench. This marking shall identify an alloy which has passed the sparking test specified in 4.6.3 and the magnetic permeability test specified in 4.6.4.

3.5 Design. Bung wrenches shall be of the double-end, all-purpose type, having multiple wrenching heads which will fit at least twenty different common commercial bungs used on storage drums, including round, square, hexagon, channel, and lug form, male and female bungs. Wrenches shall be similar to figure 1.

3.5.1 Surface finish. Forged, wrought, or cast surfaces shall be free from fins, scale, cracks, seams, burrs, or any other defect which may adversely affect the product. Flash lines shall be removed sufficiently to blend smoothly with adjacent surfaces.

3.5.2 Hardness. The heads of the wrenches shall have a Rockwell hardness of not less than 16 on the "C" scale.

3.5.3 Dimensions and weight. Wrenches shall be not less than 14-1/2 nor more than 16 inches in overall length and shall weigh not more than 3 pounds.

3.5.4 Proof loads. The heads of the wrenches shall withstand the following proof loads without permanent deformation, formation of cracks, or fracture of the heads or handles and without perceptible indentation or gouging of the bearing surfaces of the wrench heads (other than a shining of the metal):

	Inch-pounds
Small (approximately 15/16 inch) open end head.....	2,000
Small box end head.....	3,500

3.6 Workmanship. Workmanship of the wrenches furnished shall be of the quality prevailing among manufacturers normally producing wrenches of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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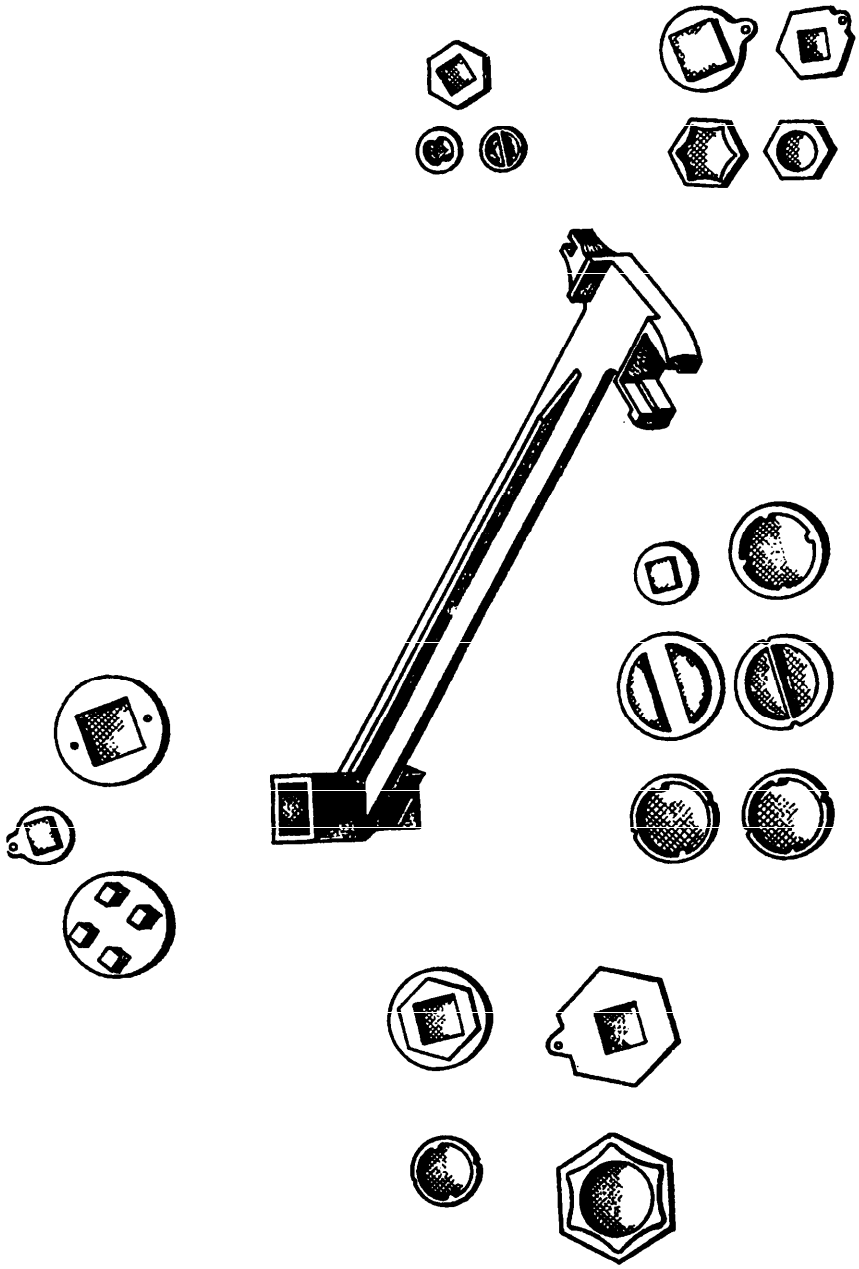


FIGURE 1. Bung wrench (showing 20 common types of bungs which wrench will fit).

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4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of all the examinations and tests required for quality conformance, the sparking test in 4.6.3, and magnetic permeability test in 4.6.4. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.4).

4.2.1 Samples for qualification tests. Three (3) sample bung wrenches shall be submitted for qualification tests.

4.3 Certification. The supplier shall furnish a certified report of the chemical composition of the material used in the fabrication of the end item certifying that the composition of each lot or batch of the material used is the same as that used in the product for which qualification was obtained. The Government reserves the right, at its option, to verify the correctness of such certificates.

4.4 Sampling for quality conformance inspection.

4.4.1 Lot. All nonsparking, nonmagnetic bung wrenches offered for delivery at one time shall be considered a lot for the purposes of quality conformance inspection.

4.4.2 Samples. A random sample of bung wrenches shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-4.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed on each sample bung wrench selected in accordance with 4.4.2. Quality conformance inspection shall consist of the following:

(a) Examination (4.5.1).
(b) Tests (4.5.2).
(c) Inspection of preparation for delivery (4.7).

4.5.1 Examination. Each of the sample bung wrenches selected in accordance with 4.4.2 shall be visually and dimensionally examined to verify conformance with requirements for marking, surface finish, weight, size, and workmanship, as specified in this specification. The acceptable quality level (AQL) shall be 2.5 percent defective. Any wrench in the sample containing one or more visual or dimensional defects shall not be offered for delivery and if the number of defective wrenches in any sample exceeds the acceptance number for the sample, the lot represented shall be rejected.

4.5.2 Tests for quality conformance. Each sample selected in accordance with 4.4.2 shall be tested for compliance with the requirements of this specification. The acceptable quality level (AQL) shall be 2.5 percent defective. Nonconforming wrenches shall not be offered for delivery and if the number of defective wrenches exceeds the acceptable number for the sample, the lot represented shall be rejected. Quality conformance tests shall consist of the tests specified in 4.6.1 and 4.6.2.

4.6 Test procedures.

4.6.1 Hardness test. The wrenches shall be subjected to a Rockwell hardness test to determine conformance with the hardness requirements of this specification (see 3.5.2).

4.6.2 Proof load test. The small open end head and the small box end head of the wrenches shall be tested by the following method: With the head of the wrench gripping a fixed mandrel of suitable shape and size to fit the opening of the head, the required proof load (see 3.5.4) shall be applied to the wrench handle through a knife edge for 10 seconds as far away from the wrench head as practicable. Test mandrels shall have a minimum Rockwell hardness of C40. The required test load shall be determined by dividing the specified proof load in inch-pounds, by the distance in inches, between the center of the mandrel and the line in which the load acts. A small "V-groove" may be cut into the wrench handle to provide a point at which the proof load can be applied. The load may be applied by means of dead weights or a suitable testing machine.

4.6.3 Sparking test.

4.6.3.1 Apparatus. The sparking test shall be conducted in an explosion chamber having a volume of 4.5 cubic feet. The chamber shall be furnished with the following: A hot plate for vaporising gasoline; a means for admitting gasoline and oxygen at

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atmospheric pressure; a fan for mixing the gasoline vapors and oxygen; two electrodes or other suitable means for providing an electric spark; a 3-inch diameter by 1/4-inch-thick metal wheel having a coarsely knurled surface on the circumference and rotating at a speed of 10,000 revolutions per minute; and a fixture for holding the test specimen against the circumference of the metal wheel while applying pressure from the outside of the chamber. The metal wheel shall be made from carbon tool steel conforming to class WI-09 of QQ-T-580 and heat treated to a Rockwell hardness of C-60 to 65. The gasoline to be used during these tests shall be aviation gasoline conforming to grade 100/130 of MIL-G-5572. An oxygen analyzer shall be used for determining the percentage of oxygen in the chamber.

4.6.3.2 Procedure. If practicable, a test specimen, approximately 1/4 inch in diameter by 2 inches long, shall be made from each questionable material used in the sample tool and shall be subjected to the sparking test. If the tool contains a previously untested, questionable material in a part too small to provide a full sized test specimen or a specimen which will provide the desired test conditions, the laboratory may request the supplier to furnish a sample of the material which is large enough to provide the required test specimen. The supplier shall certify in writing that the sample material is identical to the material used in the part of the sample tool which was to small for testing. With the specimen mounted in the holder fixture and the chamber sealed, oxygen shall be admitted into the chamber until the oxygen content in the air reaches 50 percent. A pan resting on the hot plate shall then be filled with 20 cubic centimeters of gasoline through an opening at the top of the chamber, the opening sealed, the fan turned on, and the gasoline evaporated. After the gasoline is completely evaporated, the fan shall be turned off and the specimen held against the rotating wheel with a force of 2.5 pounds. If no explosion occurs before 5 seconds, the mixture shall then be exploded by means of the electric spark to demonstrate the presence of an explosive mixture. Three runs shall be made with the test specimen and if the specimen explodes the gasoline and oxygen-enriched air mixture during any of these runs, the tool represented by the specimen tested shall be disapproved for use as a nonsparking tool.

4.6.4 Magnetic permeability test. The magnetic permeability test shall be conducted with a permeability indicator similar or equal to that specified in MIL-I-17214, according to operating instructions accompanying each indicator. The magnetic permeability shall not be above 1.2 of the indicator scale.

4.7 Inspection of preparation for delivery. Inspection of preparation for delivery shall be performed in accordance with MIL-H-15424.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders).

5.1 Preservation and packaging.

5.1.1 Level A. Preservation and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 Level C. Wrenches shall be preserved and packaged in a manner to afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice, when such meets the requirements of this level.

5.2 Packing.

5.2.1 Level A. Wrenches, packaged as specified (see 6.2), shall be packed in accordance with Level A requirements of MIL-H-15424.

5.2.2 Level B. Wrenches, packaged as specified (see 6.2), shall be packed in accordance with Level B requirements of MIL-H-15424.

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5.2.3 Level C. Wrenches, packaged as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with FED. STD. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-H-15424.

6. NOTES

6.1 Intended use. Bung wrenches covered by this specification are intended for use in the presence of flammable gases, dust and explosives, and for minesweepers, and other applications where low magnetic permeability is a factor.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Marking, if different (see 5.3).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL GGG-W-642, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

MILITARY INTEREST:

Custodians:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - GL
Navy - SH
Air Force - 84
DSAH - GS

CIVIL AGENCY INTEREST:
Preparing activity:

Navy - SH

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies of documents referenced herein. Price of this specification 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

WRENCH, BUNG, (NONSPARKING AND NONMAGNETIC) - GGG-W-642B

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4883, WHICH IS OBSOLETE
(NAVSHIPS OVPRT 13-66)

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